Sibi Vishtan Thirukonda

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Data Science grad @ Northeastern, Open to work

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EDUCATION

Northeastern University, M.S in Data Science — *Boston, MA*
Khoury College of Computer sciences, Courses: Machine Learning, Natural Language Processing, Algorithms, Time Series — *2023-Present*

Anna University, B.E in Computer Science and Engineering — *Madurai, India*
K.L.N College of Engineering, Courses: Artificial Intelligence, Operating Systems, Database Management, Distributed Systems — *2018-2022*

EXPERIENCE

Lennox International — *Chennai, India*

Individual Contributor 1 (Controls Engineering) — *Aug 2022-Dec 2022*

- **Alexa and Apple HomeKit Services:** Collaborated with cross functional team to integrate **Alexa and Apple HomeKit services** into next-generation thermostats, resulting in a ***30% increase*** in user engagement and a ***20% decrease*** in support requests.
- **Apple Watch HomeKit:** Initiated and led the development and implementation of groundbreaking features for controlling HVAC systems via Apple Watch and thecomplications, leveraging the power of Apple HomeKit APIs.
- **HVAC Stability Protocols**: Provided HVAC stability protocols using data visualization, resulting in a ***15% increase*** in system stability.
- **CMake Automation:** Achieved significant time and productivity gains by automating software modules using CMake for S30 Modules.
- Managed and maintained the internal software and alert messaging API with exceptional reliability, ensuring an impressive 99.9% uptime.

MLGround — *Bangalore, India*

Machine Learning Intern (Data Science Team) — *Jan 2022-Apr 2022*

- Conducted statistical modeling to analyze customer churn patterns and identify key factors influencing customer retention, enabling the bank to implement targeted retention strategies and reduce churn by 15% and Utilized regression analysis to determine the impact of marketing campaigns on customer acquisition, resulting in the optimization of marketing budgets and a 20% increase in new customer acquisition.
- Developed a model using time series analysis to forecast market trends and fluctuations, providing valuable risk management decisions.
- Applied clustering techniques to segment the bank's customer base into distinct groups based on their financial behavior and needs, leading to the creation of personalized product offerings and a 10% increase in customer satisfaction and conducted hypothesis testing and A/B testing to evaluate the effectiveness of different pricing strategies.

PROJECTS

Chaotic Time Series – Butterfly Effect view — *May 2023 –Present*

- Achieved an accuracy rate of 93% in predicting the behavior of chaotic time series, surpassing baseline models.
- Showcased ***state-of-the-art performance in predicting future values of the chaotic time series.***
- Utilized GPU-accelerated PyTorch and TensorFlow to train and run deep learning models for time series analysis, and LSTM model is used.

Image Generation AI/ML model (CIFAR 10) view — *Jan 2023 – Apr 2023*

- Achieved **88%** accuracy after approximately **400 epochs**, demonstrating expertise in hyperparameter tuning, model optimization and GPU acceleration for *CNN*. Further improved the model's accuracy to over **98%** using the Densenet Convolution technique, showcasing proficiency in implementing cutting-edge deep learning methods on the CIFAR-10 dataset, consisting of 60,00032 x 32 colour images.
- Showcased ability to classify images into 10 different classes, handle large datasets and leverage advanced techniques to achieve ***state-of-the-art performance in image classification and generation***.

Automation System to Limit COVID - 19 Using Face Mask Detection view — *Jan 2022 – Jul 2022*

- Developed a ***state-of-the-art Computer Vision*** Face Mask Detection system. To detect face masks in static images and in real-time streams.
- Utilized GPU accelerated Computer Vision and industry-leading frameworks, including OpenCV, TensorFlow, Caffe-based face detector, Keras, MobileNetV2, and HAAR CASCADE algorithms, Leveraged the power of these frameworks to implement robust image processing.
- Received the **Best Project award for the 2018-2022 batch at KLN College of Engineering**.

Chicago cyclist Bike Share Analysis view — *Aug 2021 – Apr 2022*

- Developed a predictive model that forecasted annual membership signups, resulting in a **20% increase** in membership acquisition.
- Leveraged dataset acquired through ***Google API*** to compare the usage patterns of casual riders and annual members of Cyclist bikes.

SKILLS

Computer Sciences: Advanced: Python, R. Intermediate: C++, SQL. **Machine Learning/Neural Networks:** SVM, CNN, DenseNet, Time Series, Physics informed Neural Networks. **Activation Functions:** ReLU, Softmax, PAF, tanh, Swish, sigmoid. **Data Analytics:** Apache, Hadoop, Pandas, NumPy, Selenium, BeautifulSoup. **Database Tools:** MySQL, MongoDB, Cassandra, UML, Entity Frameworks. **Cloud:** IBM Cloud, AWS, Google Cloud, Azure, DataBricks. **Libraries:** Spark, PyTorch, TensorFlow, scikit-learn, Caret, Shiny, Kafka

ACHIEVEMENTS AND PUBLICATIONS

- **Best Project Winner 2018-2022 Batch**: An automated system to limit COVID-19 using face mask detection.
- **Topped as Runner-up:** 72 Hour Hackathon where Kurger app was developed and recognized by Information and Technology and other departments of KLNCE.
- **IIPC Funded Competition Winner:** Won for the best project for Traffic Clearance for Ambulance and more innovative, which helps society and can be implemented in real-time at an affordable cost. And Published under TowardsDataScience
- **Best Performer:** Awarded by KLN College of Engineering as performing best in the Computer Science Department, January 2019.
- **Published Journal** under CoinsBench, Chinese Auction using Solidity EVMs.
- **Rank holder - 2:** All over Computer Science Engineering batch 2018-2022 Rank holder placed 2.
- **Published Journal** on NFC based Attendance systems under TowardsDataScience Community.
- **Published Journal** on Traffic Clearance for Ambulance under TowardsDataScience Community.
- **Worked as Research Assistant** for a Professor at Anna University helping the Data Sciences of the Patent.